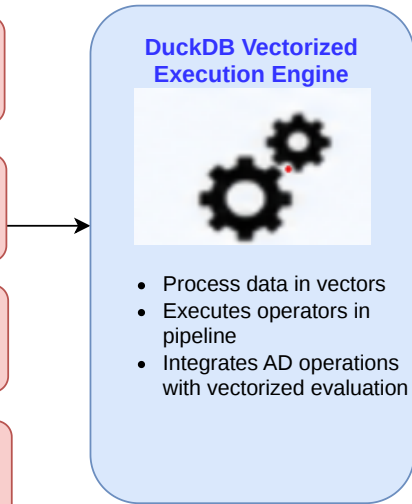
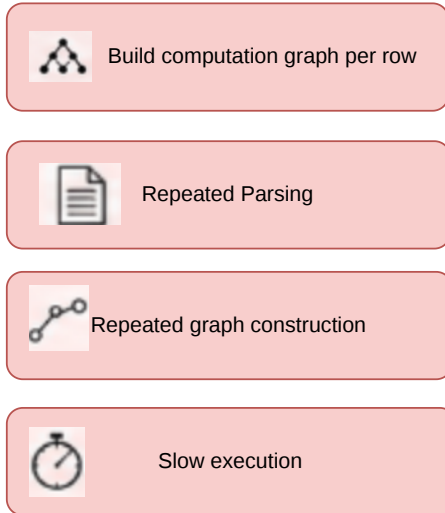


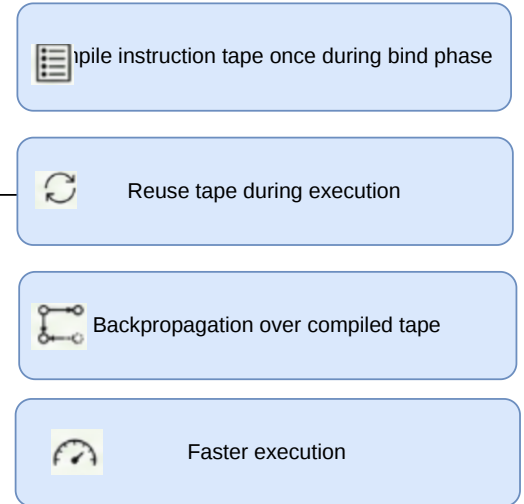
Architecture Summary and Key Takeaways

By compiling the AD computation graph into an instruction tape once-during the bind phase and reusing it in Duck DB's vectorized execution engine, we eliminate per-row overhead and achieve significantly faster reverse-mode AD.

Before Optimization (Per-Row AD)



After Optimization (Compiled Tape AD)



Key Benefits

Eliminates per-row parsing and graph construction overhead

Higher Throughput via vectorized execution

Lower CPU overhead

Scales to large datasets and complex expressions

Result: Efficient, Scalable reverse-mode AD integrated into DuckDB